SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1707**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I V.Lasya(192524186)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

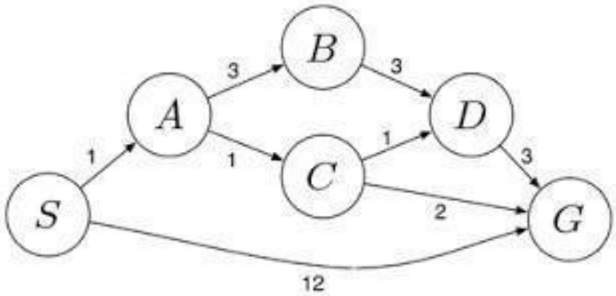
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent? ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
		requirements accurately			

RUBRICS FOR CALCULATION:

Question 1: Water Jug Problem

Problem Understanding (20%)

The Water Jug Problem is a classical Artificial Intelligence problem that demonstrates state-space search. Two jugs of 4 gallons and 3 gallons are provided without any measuring marks. The objective is to obtain exactly 2 gallons in the 4-gallon jug using only filling, emptying, and pouring operations.

Method Selection

The State Space Search method is used because each water distribution in the jugs is represented as a state. The search explores valid states until the required goal state is reached.

State Representation:

(Water in 4-gallon jug, Water in 3-gallon jug)

Solution Process

Step	State	Action
1	(0,0)	Initial State
2	(0,3)	Fill the 3-gallon jug
3	(3,0)	Pour into the 4-gallon jug
4	(3,3)	Fill the 3-gallon jug again
5	(4,2)	Pour into the 4-gallon jug until full
6	(0,2)	Empty the 4-gallon jug
7	(2,0)	Pour the remaining 2 gallons into the 4-gallon jug

Goal State: (2,0)

Accuracy of Calculation

The final state contains exactly 2 gallons in the 4-gallon jug, satisfying all given constraints. Every step follows the permitted operations correctly.

Interpretation of Results

The Water Jug Problem shows how AI solves problems using systematic search instead of trial and error. It demonstrates concepts such as state representation, search strategy, and goal testing.

Conclusion

The Water Jug Problem was successfully solved using State Space Search. The goal state (2,0) was reached through a sequence of valid operations, showing how AI applies logical reasoning to solve constraint-based problems efficiently.

Question 2: Mars Rover Intelligent Agent

Problem Understanding

The Mars Rover is an autonomous robot that explores the surface of Mars. It collects scientific data, avoids obstacles, analyzes soil and rocks, and sends information to Earth. Since communication with Earth is delayed, the rover must make intelligent decisions on its own.

Method Selection

The PEAS (Performance Measure, Environment, Actuators, Sensors) framework is used to analyze the Mars Rover because it clearly describes the components of an intelligent agent. **Solution Process**

i. Percepts

The rover receives information through:

- Camera images
- Rock and soil data
- Temperature
- Atmospheric pressure
- Obstacle sensors
- Battery level
- Navigation position

ii. Operating Environment

- Partially Observable
- Single Agent
- Sequential
- Dynamic
- Continuous
- Unknown

iii. Actions

The rover can:

- Move forward/backward
- Turn left or right
- Capture images
- Collect soil samples
- Analyze rocks
- Avoid obstacles
- Send data to Earth

iv. Performance Evaluation

The rover is evaluated based on:

- Sample collection accuracy
- Distance covered
- Battery efficiency
- Safe navigation
- Mission completion
- Data transmission quality

v. Suitable Agent Architecture

A **Utility-Based Learning Agent** is the most suitable because it learns from experience, selects the best action, saves energy, and adapts to changing conditions.

Accuracy of Analysis

The PEAS framework correctly identifies the rover's sensors, environment, actions, and performance measures. The selected architecture supports autonomous and intelligent decisionmaking.

Interpretation of Results

The Mars Rover uses AI to explore unknown environments safely and efficiently. Its intelligent decision-making improves mission success and reduces human intervention.

Conclusion

The Mars Rover is a practical example of an intelligent agent. Using the PEAS framework, its functions were analyzed effectively. A Utility-Based Learning Agent is the best choice because it supports learning, adaptation, and efficient decision-making during space exploration.

Question 3: 8-Queens Problem

Problem Understanding

The 8-Queens Problem is a classical Artificial Intelligence (AI) search problem. The objective is to place 8 queens on an 8 × 8 chessboard such that no two queens attack each other. A queen can attack another queen if they are in the same row, column, or diagonal. The goal is to find a valid arrangement where all eight queens are placed safely.

Method Selection

The most suitable method is Backtracking Search, a depth-first search technique. It places queens one by one and checks for conflicts. If a conflict occurs, the algorithm backtracks and tries another position. This reduces unnecessary searches and efficiently finds a valid solution.

Solution Process Problem

Components

Initial State:

- All possible ways of placing queens on the board.

Successor Function:

- Place one queen in the next column only if it does not attack any previously placed queen.

Goal Test:

- All 8 queens are placed without sharing the same row, column, or diagonal.

Path Cost:

- Each valid placement has equal cost.

Search Strategy

1. Place the first queen in the first column.
2. Move to the next column.
3. Check for row and diagonal conflicts.
4. If conflict exists, try another row.
5. If no row is valid, backtrack to the previous column.
6. Continue until all eight queens are placed safely.

Accuracy of Solution

The backtracking algorithm ensures that every queen is placed in a safe position. It guarantees a correct solution by removing invalid placements before continuing the search.

Interpretation of Results

The 8-Queens Problem demonstrates how AI solves Constraint Satisfaction Problems (CSPs) using systematic search. Backtracking reduces the search space and provides an efficient solution compared to checking all possible arrangements.

Conclusion

The 8-Queens Problem was successfully formulated using Backtracking Search. By checking constraints at every step and backtracking whenever necessary, a valid arrangement of eight queens was obtained. This problem highlights the importance of search algorithms and constraint satisfaction techniques in Artificial Intelligence.

Question 4: OLA Cab Booking Problem

Problem Understanding

A customer wants to travel from one location to another using the OLA Cab Booking application. The application provides different cab options such as Micro, Mini, Sedan, Prime, and Shared. The customer selects a cab based on comfort, fare, and availability. The system must choose the most suitable cab and complete the booking process.

Method Selection

The most suitable problem-solving agent is a Utility-Based Agent. This agent compares different cab options using factors such as price, travel time, comfort, and availability, then selects the option that provides the highest utility (best overall benefit).

Solution Process

Pseudocode

```
START
Input pickup location
Input destination
Display available cab types
Customer selects preferred cab
Check cab availability
IF cab is available THEN
    Calculate fare
    Estimate travel time
    Display booking details
    Confirm booking
ELSE
    Suggest another available cab
ENDIF
Assign driver
Track cab location
Complete trip
Generate payment receipt
END
```

Working of the Agent

1. The customer enters the pickup and destination locations.
2. The system displays available cab types.
3. The Utility-Based Agent compares all options.
4. It selects the best available cab based on user preference.
5. After confirmation, a driver is assigned and the trip begins.

Accuracy of Solution

The Utility-Based Agent selects the most suitable cab by considering multiple factors instead of making a random choice. This improves efficiency and customer satisfaction.

Interpretation of Results

Using a Utility-Based Agent helps the OLA application provide better recommendations by balancing cost, comfort, and travel time. It enables intelligent decision-making and enhances the overall user experience.

Conclusion

The OLA Cab Booking system is best represented by a Utility-Based Agent because it evaluates multiple alternatives before selecting the best option. This approach improves customer satisfaction, optimizes resource usage, and demonstrates the practical application of Artificial Intelligence in real-world transportation systems.

Question 5: Uniform Cost Search (UCS)

Problem Understanding

A logistics company needs to transport packages from the starting warehouse S to the destination warehouse **G** with the minimum travel cost. Each route between warehouses has an associated cost, which includes fuel and travel time. The objective is to find the least-cost path using the Uniform Cost Search (UCS) algorithm.

Method Selection

The most suitable algorithm is Uniform Cost Search (UCS) because it always expands the node with the lowest cumulative path cost. UCS guarantees the optimal solution when all edge costs are non-negative, making it ideal for logistics and route planning.

Solution Process

Sample Weighted Graph

Assume the graph contains the following routes:

Route	**Cost**
S → A	2
S → B	5
A → C	4
A → D	7

B → E	3
C → G	3
D → G	2
E → G	6

UCS Steps

1. Start from node **S** with cost **0**.
2. Expand the node with the lowest cumulative cost.
3. Add neighboring nodes to the priority queue.
4. Continue expanding the lowest-cost node.
5. Stop when the destination **G** is reached.

Path Cost Calculation Path

1:

S → A → C → G

Total Cost = 2 + 4 + 3 = 9 **Path**

2:

S → A → D → G

Total Cost = 2 + 7 + 2 = 11

Path 3:

S → B → E → G

Total Cost = 5 + 3 + 6 = 14

Optimal Path

S → A → C → G

Minimum Cost = 9

Accuracy of Calculation

The total cost of each possible path is calculated correctly.

- Path 1 = **9**
- Path 2 = **11**
- Path 3 = **14**

Since **9** is the lowest cost, **Uniform Cost Search** correctly selects **S → A → C → G** as the optimal path.

Interpretation of Results

Uniform Cost Search efficiently finds the least-cost path by expanding the node with the minimum cumulative cost. This approach helps logistics companies reduce transportation costs, save fuel, minimize delivery time, and improve overall operational efficiency.

Conclusion

The logistics routing problem was successfully solved using the Uniform Cost Search (UCS) algorithm. UCS guarantees the optimal solution by always selecting the path with the lowest accumulated cost. The optimal route **S → A → C → G** with a total cost of 9 was identified as the best path. This algorithm is widely used in AI applications such as GPS navigation, robotics, network routing, and supply chain management because of its accuracy and efficiency